Gaurav Lamba

ic23btech11007@iith.ac.in | gauravlamba714@gmail.com | +91 7082743108 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.42
RPS Public School	Class XII (CBSE)	2022	93.80%
RRCM Sr. Sec. School	Class X (CBSE)	2020	94.60%

SCHOLASTIC ACHIEVEMENTS

Among the top 1% of over 1.2 million candidates registered for JEE 2023.

NTSE Scholar 2020 : Awarded to **1,000 students nationwide** out of 1.2-1.4M candidates, recognizing exceptional academic excellence and analytical aptitude.

WORK EXPERIENCE

AI Engineer Intern | Tredence Inc.

Awarded 'Best Intern Project & Most Practical Solution' in India-wide Intern Ideathon

- Built end-to-end RAG data pipelines for structured and unstructured ingestion (PostgreSQL, pgvector); architected **MAC-SQL**, a multi-agent SQL synthesis system via LangGraph, grounded in 8+ **BIRD**-benchmarked research papers.
- Shipped production solutions across Healthcare, BFSI, and Retail -- demoed directly to **Google and eBay** -- via deterministic DAG workflows with conditional routing, state passing, and HITL.
- Built a hybrid synthetic data generation framework (LLMs + **SDV/TVAE**) enforcing referential integrity via Kahn's-topological-sort-based foreign key injection.
- Fine-tuned a 7B LLM (**LoRA** 4-bit quantization, **GRPO** reinforcement learning) and deployed a self-correcting execution loop natively on Ollama for optimized local inference.
- Shipped reusable agent patterns for new verticals and a local, runnable Text-to-SQL pipeline.

PROJECTS

Sandboxed Code Execution & Judging Platform

- Docker-sandboxed code execution engine in Go with per-submission CPU, memory, and network limits.
- PostgreSQL job queue with row-level locking and heartbeat-based failure recovery across concurrent workers.
- Multi-language judging engine (C++, Python, Java) with automated verdict classification per test case.

Typo-Tolerant Autocomplete & Search Engine

- Trie-based autocomplete engine with min-heap top-k ranking for fast prefix search over a 100K+ entry dataset.
- Custom Levenshtein edit-distance algorithm wrapped in a BK-tree for typo-tolerant fuzzy matching.
- Latency benchmarking against naive linear search to quantify query-speed gains at scale.

AI-Powered Document Processing System

- Automated extraction of data from PDFs and handwritten images using Python.
- Integrated Azure OpenAI, Google Gemini, and Groq APIs for text and vision analysis.
- Produced structured outputs in fixed and dynamic CSV schemas for scalable workflows.

Generative Molecular Design Pipeline (CVAE) for Materials Discovery | *Prof. Nagarajan Ganapathy*

- End-to-end generative ML pipeline (SMOTE-SELFIES-CVAE-Constrained generation) for inverse molecular design.
- Conditional VAE posterior collapse diagnosed and resolved via cyclic KL annealing and free-bits regularization.
- Random Forest property predictor (Morgan fingerprints, LOO-CV) achieving ~15% mean relative error, generating 12 target-matched candidate structures on a 52-sample dataset preprocessed to 500.

SKILLS

Programming Languages: C, C++, Python, Go, SQL

Development & Tools: Git, Docker, PostgreSQL, Jupyter/Colab, PyTorch, TensorFlow/Keras, Scikit-Learn

Core Competencies: Data Structures & Algorithms, Operating Systems, OOPs, DBMS, Computer Networks

Machine Learning: Deep Learning, Neural Networks (ANN, CNN), GenAI with LLM, Transformers, Agentic AI

Soft Skills: Effective Communication & Collaboration, Problem-Solving, Leadership, Critical Thinking, Time Management

RELEVANT COURSES

CSE Electives: Introduction to Artificial Intelligence(ID1050), Introduction to Programming(ID1063)

Mathematics Elective: Calculus(MA1110, MA1220), Probability(MA2110), Elementary Linear Algebra(MA1140), Differential Equations(MA1150)

Departmental Core: Machine Learning in Chemistry(IC3023)

POSITIONS OF RESPONSIBILITY

Overall Head - BTL, The Filmmaking Club (2025-2026): Spearheading all club operations, leading a cross-functional team in **event planning, production management, and creative direction** for campus filmmaking projects and competitions.

Assistant Manager - Operations, E-Cell, IIT Hyderabad (2023-2024): Managed end-to-end **operations for E-Summit**, the institute's flagship entrepreneurship festival, ensuring seamless execution of 20+ events, coordinating with 50+ startups & speakers, and optimizing on-ground logistics to handle 5,000+ participants.

EXTRACURRICULAR

Participant - Filmmaking Competition, Inter IIT Cultural Meet 7.0:

Represented IIT Hyderabad in the national-level cultural festival, collaborating in a team to conceptualize, script, and produce a short film under competitive timelines.

NSS: Volunteered in National Service Scheme events.